THE HUMAN BATTERY™ PROJECT

INTERMEDIATE

NINETY-DAY PROTOCOL

You know what you should be doing and you do some of it. This turns sometimes into every day.

THE MODEL

Trillions of batteries, keeping time together

Your body is not one battery. It is a coordinated network of trillions of cellular batteries. Every cell holds a voltage across a membrane, and different tissues are built to hold different voltages. A neuron, a muscle cell and an immune cell are supposed to be different. The goal is the voltage appropriate to each tissue, never the maximum.

Coherence is the ability of cells to hold their proper electrical identity while coordinating their voltage, metabolism, timing and behavior with the tissue around them and with the whole organism. A cell can be individually healthy and still be out of step with its neighbours. That is what the ninety days are for.

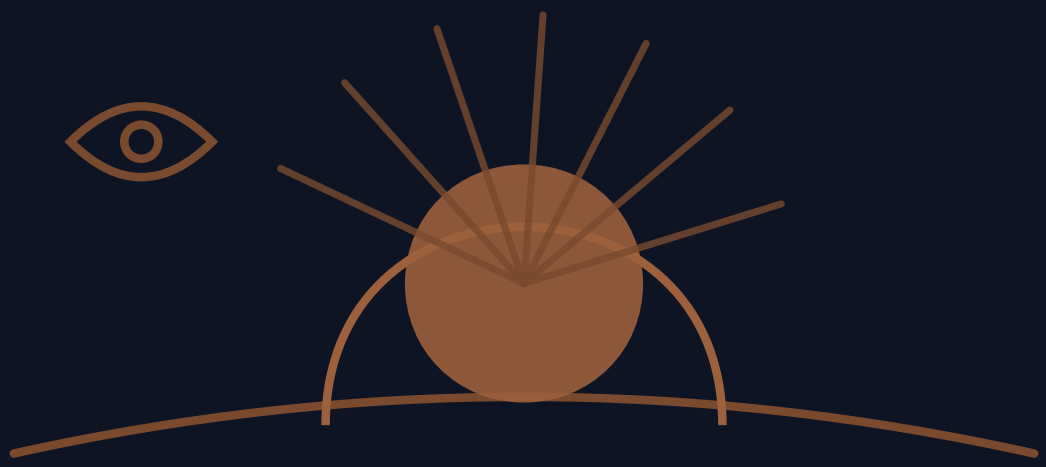
Light does two jobs. The big one is timing. Morning light tells the body what hour it is, and that signal sets the nervous system, hormones, metabolism and mitochondrial activity. The small one is chemistry in your skin: sunlight makes vitamin D and releases nitric oxide, which widens your arteries. So light is a real energy input, and a small one. Your skin takes in a lot of sunlight and nearly all of it becomes heat, because the body cannot turn light into fuel.

Eating with the season is in the protocol because different foods give your body different fuel, and daylength sets how your body handles fuel. Lining those up is the idea being tested. It is not a settled fact, and food does not carry any message about the month it grew in.

The circadian protocol is how coherence is supported. The daily log is how it is tracked. Every element in this document carries a note saying which of the four questions it answers, and where an element answers none of them, it says so.

Circadian protocol

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.



LIGHT SETS THE CLOCK

1

01 / THE CLOCK

Circadian protocol

INTERMEDIATE

Everything in this program depends on this section. If you do nothing else in the first two weeks, do this.

Your body runs on a clock, and the clock is set by light. Morning light tells every cell what time it is. Darkness at night lets the repair cycle run. When the clock is right, food is handled correctly, hormones release on schedule, and sleep does its job. When the clock is wrong, nothing downstream works, no matter how well you eat or train.

The sequence, every day:

LIGHT → WATER → MOVEMENT → FOOD → MOVEMENT → LIGHT → DARKNESS

Go outside as soon as you wake up

What to do: Outside within 30 minutes of waking. Fifteen minutes minimum, twenty when cloudy. Barefoot on the ground when you can. Face east. No sunglasses.

Why: Your body has a clock inside it. The clock does not know what time it is until it sees the sun. When morning light hits your eyes, it tells every cell in your body that the day has started. That one signal sets up your energy, your hunger, your hormones and your sleep for the whole day. If you skip it, your body runs on the wrong time all day, like a clock that was never set.

Barefoot if you can. This is part of the morning light block, not separate from it. We include it because it costs nothing, and because standing on grass keeps people outside for the full time when shoes and a doormat do not. We are not measuring it, and we are not claiming it does anything on its own.

> **Answers Q3, Q4.** Sets the circadian clock every tissue runs on. Sleep timing and light exposure are logged daily.

Get sun on your skin in the afternoon

What to do: Fifteen minutes of sun on your arms and face in the afternoon. Build up slowly if your skin is fair.

Why: Morning light sets your clock. Midday sun holds it there. It is also the only time of day your skin can make vitamin D, and sun on bare skin widens your blood vessels, which lowers blood pressure. We measure both at day 0 and day 90.

> **Answers Q3, Q4.** Holds the clock set. Vitamin D is on the panel and blood pressure is measured at both ends.

Watch the sun go down

What to do: Outside for the last fifteen minutes of light.

Why: The light at sunset is red and orange. That color tells your body the day is ending and it is time to start making melatonin, the hormone that makes you sleepy. Seeing the sunset is how your body knows to begin winding down. Skip it and your body does not get the message, and then you are wide awake at midnight wondering why.

> **Answers Q3.** Red light starts melatonin onset, the handover from day signaling to night signaling.

Put on your glasses when the sun goes down

What to do: Amber lenses from sunset. Bedroom and living room switched to incandescent or red bulbs. Iris on every screen. Film on the TV.

Why: After sunset, your body expects darkness. Light bulbs and screens give off a blue light that looks like daytime to your brain. Your brain sees it and stops making melatonin, and that is why you cannot fall asleep even when you are tired. Amber or red glasses block the blue. So do warm bulbs and screen filters. It is not about the brightness. It is about the color.

> **Answers Q3.** Protects that melatonin signal from blue light that would tell the body it is still daytime.

Make the bedroom dark and quiet

What to do: Phone out of the bedroom. Router on a timer, off 10pm to 6am.

Why: Your body does its repair work while you sleep, and it can only do it in the dark. Even a small light, like a phone screen or a charger glow, can be enough to keep the repair from starting. The phone also keeps your brain on alert, waiting for a buzz. Out of the room means your body can finally switch off.

> **Answers Q3, Q4.** Darkness is when repair signaling runs. Sleep timing and regularity are logged daily.

Give your body hours with no food

What to do: Eating window 8am to 5pm. A 15-hour fast every night. One 24-hour fast per month once the window is easy.

Why: Every time you eat, your body has to stop what it is doing and deal with the food. If you eat from the minute you wake up until right before bed, your body never gets a break to do repairs. Hours without food, especially overnight, is when the cleanup happens. Late eating is the worst, because your body is trying to wind down and you just handed it a job.

> **Answers Q1, Q4.** Metabolic flexibility and substrate switching. Fasting glucose, insulin and HbA1c are on the panel.

> **The rule that matters most:** light before food, every single day. Do not eat until you have been outside.

02 / WATER

Water and minerals

The morning glass first. Then three liters, every bottle with minerals in it.



MINERALS, LEMON, SALT

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02 / WATER

Water and minerals

INTERMEDIATE

The morning glass

Before coffee, before food, before anything: 16 to 24 oz of mineral water with a pinch of unrefined mineral sea salt and 1 to 2 oz of pure organic lemon juice. Empty stomach. It is the first thing the body receives after light.

Through the day

Three liters minimum, more on training days. Every bottle gets a pinch of unrefined mineral sea salt, because water without minerals passes through and takes minerals with it. Most of it before your last meal.

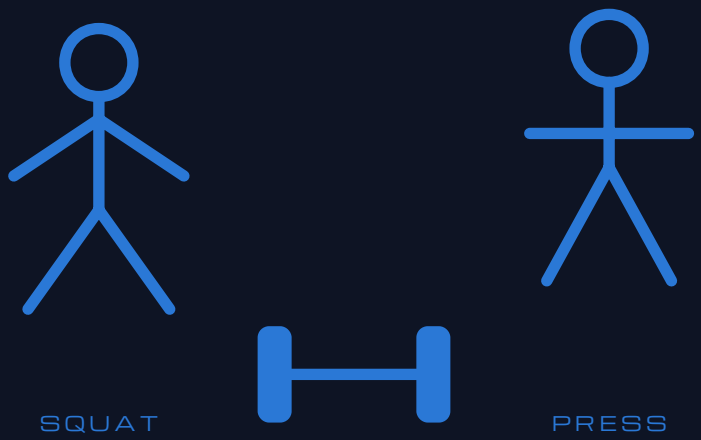
What to drink

Bottled water with a measurable mineral content, not distilled. At home, reverse osmosis with unrefined mineral sea salt added back, a quarter teaspoon per liter, and the same salt for cooking. Nothing else. We take no commission on anything named in this guide.

> **Answers Q2, Q4.** Minerals are the ions the pumps move. Sodium, potassium and RBC magnesium are on the panel.

Movement

Four patterns. Walks after every meal. Zone 2 for the mitochondria, intervals for the ceiling.



SQUAT, PUSH, PULL, HINGE

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Movement

INTERMEDIATE

Resistance, three days

Bodyweight progressions. **Squat:** bodyweight to goblet to Bulgarian split squat. **Push:** incline push-up to floor to dips. **Pull:** band rows to inverted rows to pull-ups. **Hinge:** band hinge to kettlebell Romanian deadlift. Three sets of eight to twelve. Progress when you clear twelve on all three.

Zone 2, three sessions of 40 minutes

Brisk walking, incline treadmill, easy cycling. You can talk.

Intervals, once a week

30-20-10: thirty seconds easy, twenty moderate, ten hard, five times through. Rest two minutes. Repeat the block. Twenty minutes total.

Walking

Ten minutes after every meal, fifteen after dinner. Eight thousand steps.

Grip

Dead hangs, three sets to failure, twice a week.

> **Answers Q1, Q4.** Builds mitochondrial number and respiratory capacity. VO2max, lactate and grip strength are measured at both ends.

Food

Protein first, fat for fuel, inside a window that closes early. The full list is in the Dietary Guidelines.



PROTEIN, FAT, THEN CARBS

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04 / FOOD

Food

INTERMEDIATE

Window: 8am to 5pm

Three meals or two. No snacks.

First meal

Three pastured eggs, or a tin of sardines four days a week. Leafy greens. A forkful of sauerkraut or kimchi. Olive oil or butter.

Midday

6 to 8 oz protein: beef, chicken thigh, salmon, lamb. Two cups vegetables, half cruciferous. Sweet potato or white rice, one fist. Olive oil.

Last meal, by 5pm

6 oz protein. Vegetables. Avocado.

Every day

Eggs or sardines every morning. Greens every day. Fermented food every day. Salmon or sardines four times a week.

Protein

1.6 grams per kilogram.

Out for 90 days

Gluten, seed oils, added sugar, alcohol. Grains limited to white rice and sweet potato. Nightshades stay in.

> **Answers Q1, Q2, Q4.** Fuel handling, plus the fatty acids and protein the membranes are built from. Glucose, insulin, HbA1c, Omega-3 Index and albumin are on the panel.

The full approved food list, the daily non-negotiables and how to build a plate are in the Dietary Guidelines document.

Heat and cold

Stress that builds capacity. Dosed to your tier, timed away from your lifts.



HEAT COLD

STRESS THAT BUILDS CAPACITY

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Heat and cold

INTERMEDIATE

Heat, three times a week

Fifteen minutes at about 80C (176F).

Cold, one of two ways

Sixty to ninety seconds at the end of every shower, daily. Or a one to two minute plunge at 12 to 15C (54 to 59F), two to three times a week. Pick one and hold it for the cycle rather than alternating.

> **Cold is never within six hours after strength training.** Cold before a lift is fine. Cold after a lift blunts what the lift was for. If you train in the morning, cold goes in the evening or on a day you did not lift.

> **The sauna finishes one to two hours before bed.** Later than that and the heat is still leaving your body when you are trying to fall asleep.

> **Two limits that never move.** No single cold exposure over five minutes, and no water colder than 7C (45F). These hold at every tier, and they hold when your intensity rises between cycles. Colder and longer is not the direction this goes.

> **No alcohol before or during the sauna.** Mineral water after.

> **Never plunge alone in open water. Never put your head under, and never hold your breath.** The gasp reflex is involuntary and it is what makes cold water drown people who can swim.

Who skips this

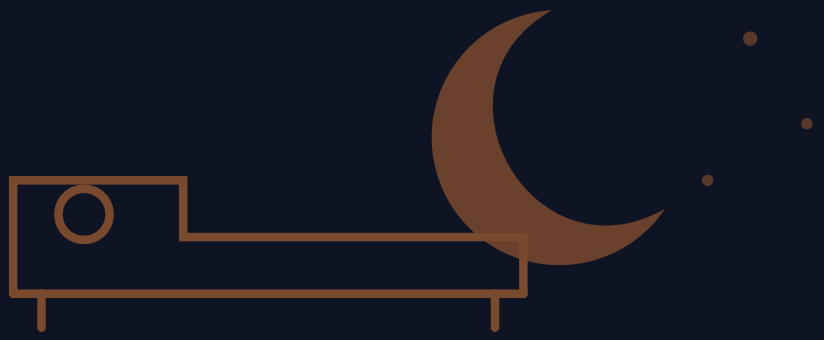
Ask your prescriber first, and treat the answer as the instruction, if any of these is true of you: a heart attack in the last twelve months, unstable chest pain, severe aortic stenosis, an arrhythmia, blood pressure that is not controlled, very low blood pressure, or pregnancy. **Cold specifically:** Raynaud's. **Heat specifically:** if you are a man trying to conceive, halve the dose rather than removing it.

What we are standing on. Sauna frequency and duration: *We are confident*, and the evidence is observational, which means it shows people who use a sauna more often do better and cannot prove the sauna is why. Cold exposure: *Early evidence*. The doses below are the protocol's, not a finding.

> **Answers Q1.** Controlled stress that builds mitochondrial capacity. Nothing on our panel isolates it.

Sleep

Same time, every night. Dark, cool, nothing plugged in.



DARK, COOL, SAME TIME

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06 / SLEEP

Sleep

INTERMEDIATE

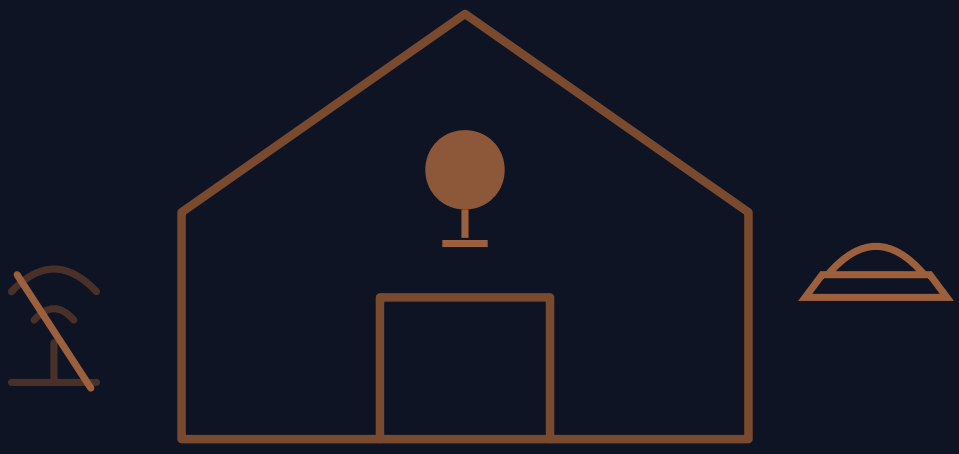
Same bedtime, same wake time, seven days a week. Room cool and completely dark. Last food three hours before bed. Phone out of the bedroom.

Fixed wind-down at 9pm: glasses on, screens off, magnesium taken.

> **Answers Q3, Q4.** Recovery is when gradients are rebuilt and membranes repaired. Sleep timing and regularity are logged daily.

Equipment and sourcing

Glasses, bulbs, a router timer, water, and where to get each one.



RED LIGHT, ROUTER OFF, GLASSES ON

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07 / ENVIRONMENT

Equipment and sourcing

INTERMEDIATE

Dr. Pittman takes no commission on anything listed here. These are what he uses.

Specifications, not brands. Each row says what the thing has to do, so you can judge any product against it yourself. No brand is named in this table. **We take no commission on anything named in this guide, and nothing here is a paid placement.** Evidence tier for this whole table: **Dr. Micah's idea, being tested**, except the rows marked otherwise.

Evening glasses	Lenses that block 460 to 490 nm. Ask any seller for the transmission spectrum and do not buy without it, because "blue light blocking" is unregulated and most amber lenses do not reach 490 nm. Red lenses block more than amber. *Tier: early evidence for the effect on melatonin onset.*
Screens	Any color-temperature shifter set to remove blue after sunset. The free ones work. A film on the monitor does the same job in hardware if you prefer not to run software.
Bulbs	Incandescent wherever you can still find them, because they have almost no blue. Otherwise a red or amber bulb in the rooms you use after dark.
Router	Mechanical outlet timer, off 10pm to 6am. The cheapest item on this list.
Grounding, optional	**Dr. Micah's idea, being tested.** Not a required part of the protocol and the evidence for it is weak. If you want to try it, a conductive sheet for the bed or a mat for the desk is the usual form. Test the outlet with an inexpensive outlet tester first. A participant who never grounds is fully adherent.
Water	Mineral water with a measurable mineral content, not distilled or reverse-osmosis alone. At home, under-sink reverse osmosis with salt added back, a quarter teaspoon per liter. *The specification for the salt: an unrefined salt with a full trace-mineral profile, not refined sodium chloride. Unrefined salts vary and most table salt does not qualify, so read the mineral analysis rather than the front of the packet.*
Sauna	Traditional, not infrared. A gym sauna is fine and is what most people should use. *Tier: strong for cardiovascular outcomes in the Finnish cohort data.*
Cold	A cold shower is enough to start. A chest freezer with a GFCI is the cheap home plunge. *Tier: contested. One unreplicated trial, and the warm-shower arm improved about as much.*
Butter	Grass-fed, any brand.
Testing	A dried-blood-spot Omega-3 Index kit at day 0 and day 90, from a lab that reports EPA+DHA as a percentage of red cell membrane fatty acids. A hand dynamometer for grip. *Tier: established for the Omega-3 Index as a defined measurement.*

Three phases

Stop the drain. Recharge. Build. Measured at both ends.



MEASURED AT BOTH ENDS

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08 / THE NINETY DAYS

Three phases

INTERMEDIATE

Days 1 to 30	Stop the drain. Morning light, fixed wake time, no snacks, last food three hours before bed. Glasses on in the evening.
Days 31 to 60	Recharge. Window to 8am to 5pm. Three resistance days. Cold finish on every shower.
Days 61 to 90	Build. Intervals once a week. First cold plunge. Day 90 draw.

Daily checklist

This is what goes in the log. Under a minute.



UNDER A MINUTE, EVERY DAY

9

09 / EVERY DAY

Daily checklist

INTERMEDIATE

Morning daylight

- [ ] Outside within 30 min of waking, 10+ min
- [ ] Midday sun 10 min

Hydration

- [ ] Morning glass
- [ ] Three liters of water

Movement

- [ ] Training or zone 2 or rest day
- [ ] Walk after each meal

Food timing

- [ ] Meals inside 8am to 5pm, no snacks

Heat and cold

- [ ] Cold finish on shower, 60 to 90 s
- [ ] Sauna if it is a sauna day, finished 1 to 2 h before bed

Sleep

- [ ] Wind-down at 9pm

Nighttime darkness

- [ ] Amber glasses two hours before bed
- [ ] Phone out of room
- [ ] Logged

THE HONEST PART

What this is, and is not

The Human Battery Project is an educational wellness program, not medical treatment. It does not diagnose or treat any condition and does not replace your physician.

Any laboratory result outside the reference range is referred to a physician. Every time.

If you take prescription medication, review any change to your eating window, and any sauna or cold exposure, with your prescriber before starting. Never adjust a medication on your own because of anything in this document.

Stop and seek medical care for chest pain, fainting, shortness of breath at rest, or any new symptom that concerns you.

Never manipulate potassium or other electrolytes to try to change your body's charge. Balance and correction of a genuine deficiency only.

> **We do not promise outcomes.** We measure at both ends and show you exactly what changed.